POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Order and Record

Product Standard No.

Product Code No.

Manufacturing Instruction No.

PMF-442

P40327

ADM-180801

Product Name

Appearance

Cephorin-F Tablet

Green oblong film-coated layered tablet

Manufacturing Number

Manufacturing Unit

Manufacturing Date

Expiration Date

18c005

200,000 tablets

20 18 . 08 . 08 .

20 20 . 08 . 07 .

Deputy Quality Assurance Manager

Manufacturing Instruction Site

Issuer

Verifier

Manufacturing Manager

Moon

2016.08.09.

Headquarters
2018. 08. 09

2018.08.09

2018.08.09

2018.08.09

※ Note (Comment)
1. This product is a commissioned product of Ahn-Gook Pharm.
2. From Poonglim's weighing date to bulk packaging, the work is completed within 30 days. (2018. 08.10. ~2018.09.08.)

None

Rim
Pre-Operation Check

Step

Verifier
Verification Date

Inspector
Inspection Date

09. 07 ~ 09. 26

Cheongryang
Preparation

Inspection Time (Start~End)

Lee Seung-Dong
08/10

JY
08/10

Moon 2 Do: 4)T
<) Raw Material Weighing Room (0814) SE (2045T), Humidity (65% or less) _Main
[OCR][4.|%

② Check for Raw Material Proper Label Attachment □ Attached □ Not Attached

③ Check Weighing Scale Leveling and Zero Point □ Suitable □ Unsuitable

④ Verify Cleanliness and Clean State of Operation Room □ Suitable □ Unsuitable

⑤ Confirm Washing, Drying, and Abnormality of Machinery and Equipment
1) Weighing Container (PRE-210) □ Suitable □ Unsuitable
2) Electronic Direct Display Scale (PRE-107) □ Suitable □ Unsuitable
3) Electronic Direct Display Scale (PRE-113) □ Suitable □ Unsuitable
4) Electronic Direct Display Scale (PRE-117) □ Suitable □ Unsuitable

2017.12.15 - Ver.08

2/70

Form No.: FP-2-01-001-01(10)

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Manufacturing Process Name

Manufacturing Unit

Manufacturing Number

Product Name

Product Code No.

18005

(Middle Layer) Mixing

200,000 tablets

Cephorin-F Tablet

P40327

Verifier
Verification Date

Inspector
Inspection Date

Pre-Operation Check

Step

Ahn Joo-Sang
08/13

11:16 ~ 11:33

Inspection Time (Start~End)

① Check Weighing Scale Leveling and Zero Point

② Re-Verify Weighed Raw Material Weight, Input and Label Record Attachment

□ Suitable □ Unsuitable

③ Confirm Washing, Drying, and Abnormality of Machinery and Equipment
1) SUS Container (40ℓ) (PRE-212) □ Suitable ☑ Unsuitable
2) Air Mixer(4) (PRM-196) ☑ Suitable □ Unsuitable
3) High-Speed Mixer(3) (PRM-225) ☑ Suitable □ Unsuitable
4) Fluidized Bed Coater (GPCG60) (PRM-107) ☑ Suitable □ Unsuitable
5) 20㎛ product retaining filter (PRE-248) □ Suitable ☑ Unsuitable
6) Wet Granulator (12, 30 mesh) (PRM-172) □ Suitable ☑ Unsuitable
7) Mobile Powder Transfer System(2) (PRM-178) □ Suitable ☑ Unsuitable

8) Sieve Mesh (30 mesh) (PRE-211) □ Suitable ☑ Unsuitable
9) U-Type Mixer (PRM-115) □ Suitable ☑ Unsuitable
10) Mobile Powder Transfer System (PRM-115-①) □ Suitable ☑ Unsuitable
④ Attach Cleaning Confirmation for Machinery ☑ Attached □ Not Attached

⑤ Verify Cleanliness and Clean State of Operation Room

□ Suitable □ Unsuitable

Worker

Verifier

Record Details

Work Process

Step

Work Date

Verification Date

Ahn Joo-Sang
08/13

Temperature: 21.9 ℃ Humidity: 54.9 %

Verify and Record Temperature (20±5℃) and Humidity (65% or less) in Mixing Room 8 (PR227)

JS
08/13

Temperature: 21.3 ℃
Humidity: 60.7 %

Park Cheon-Mok
08/13

Verify and Record Temperature (20±5℃) and Humidity (65% or less) in Granulation Room 2 (PR36)

Inspection Details

Temperature: 21.5 ℃
Humidity: 56.2 %

Ahn Joo-Sang
08/13

Verify and Record Temperature (20±5℃) and Humidity (65% or less) in Mixing Room 3 (PR63)

Form No.: FP-2-01-001-01(10)

2017.12.15 - Ver.08

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Manufacturing Process Name

Manufacturing Unit

Manufacturing Number

Product Name

Product Code No.

(Middle Layer) Mixing

18 005

200,000 tablets

P40327

Cephorin-F Tablet

Verifier

Worker

Record Details

Work Process

Step

Work Date

Verification Date

Middle Layer Mixing Manufacturing Process

08/13

Work Time (Start~End)
11:35 ~ 11:59
4,000 g
22,000 g
11:38 ~ 11:58
□ Suitable ☑ Unsuitable

Man Joo-Sang
Aug/13

Add raw materials into 40ℓ SUS Container in the following order to prepare binding solution with Air Mixer(4)

Hydroxypropylcellulose ... 4,000g
Isopropanol ... 22,000g

Stirring Time ... 20±5 mins

Visually confirm if the raw materials are dissolved

1
Binding Solution
Preparation

Work Time (Start~End)
12:00 ~ 12:06
33,600 g
24,800 g
90 RPM
1,200 RPM
12:03 ~ 12:06

Ahn Joo-Sang
08/13

Mix the following raw materials in a High-Speed Mixer (3)
Ranitidine Hydrochloride...33,600g
Microcrystalline Cellulose 101...24,800g
Agitator Speed ...90±10 RPM
Chopper Speed ...1,200±100 RPM
Mixing Time ... 3 mins

2
Mixing

Add binding solution to the above mixture and amalgamate under the following conditions
Agitator Speed ...90±10 RPM
Chopper Speed ...1,200±100 RPM
Mixing Time ... 1 min

Dan Joo-Sang
08/13

Work Time (Start~End)
12:06 ~ 12:15
90 RPM
1,200 RPM
12:10 ~ 12:11

3
Amalgamation

JS
08/13

Work Time (Start~End)
12:18 - 12:37
☑ Suitable ☐ Unsuitable
1,500 ∨ 2,800 ㎥/h
65.0 ℃
43.2 ℃
12:20 - 12:35

Jinwook Park
08/13

After checking the mesh screen, dry the composite in the fluid bed coater under the following conditions.

Verify mesh screen installation and check for any abnormalities.

4

Drying I

1,500~2,800 m³/h

Air Flow...

65.0±5.0℃

Inlet Temperature

45.0±5.0℃

Product Temperature

15±5 min

Drying Time

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POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Manufacturing Process Name

Manufacturing Unit

Manufacturing Number

Product Name

Product Code No.

(ChoongChoong) Mixing

18005

200,000 tablets

Cephorin-F Tablet

P40327

Worker

Verifier

Record Details

Work Process

Step

Work Date

Verification Date

JS
08/13

Chanwook Park
08/13

Working Time (Start~End)
12:38 - 12:52
12 mesh
☑ Suitable ☐ Unsuitable
45 RPM

After verifying the granulation mesh, feed the dried material into the wet granulator under the following conditions.

Verify granulation mesh installation and specifications...12 mesh

Check exterior and any abnormalities of granulation mesh

Rotation Speed...45±5 RPM

5
Granulator

Working Time (Start~End)
12:54 - 13:15
☑ Suitable ☐ Unsuitable
1,500～2,800 m3/h
66.3 °C
46.7 °C
13 min
Result: 0.6%

JS
08/13

Chanok Park
08/13

Verify mesh screen installation and check for any abnormalities.
Air Flow 1,500~2,800 m3/h
Inlet Temperature 65.0±5.0°C
Product Temperature 45.0±5.0°C
Drying Time 10±5 min

Dry in the machine

6

Drying II

and

In case of, end the drying process
Halogen Moisture Analyzer, 80℃, 4 min, 5g usage
Moisture Content Standard below 2.0%

Moisture Measurement

4:00 min
80 °C
5.012 g
0.00 - 100 %MC

Timer
Drying temp.
Weight

-0.18 %MC

1:00 min

---13.08.18---13:15---

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14/70

2017.12.15 - Ver.08

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Product Code No.

Manufacturing Unit

Product Name

Manufacturing Number

Manufacturing Process Name

18005

P40327

(Middle Layer) Mixing

Cephorin-F Tablet

200,000 tablets

Worker

Verifier

Step

Work Process

Record Details

Work Date

Verification Date

7
Granulation II

After verifying the granulation mesh, use the mobile powder feeder (2) to feed the dried material into the wet granulator under the following conditions.

Verify granulation mesh installation and specifications... 30 mesh

Check exterior and any abnormalities of granulation mesh

Rotation Speed... 45±5 RPM

Working Time (Start~End)
13:18 ~ 13:48
30 mesh
Suitable Unsuitable
45 RPM

Jinok Park
08/13

JS
08/13

Add the above granulated material and the following raw materials into the mixer and mix under the following conditions.
Croscarmellose Sodium 4,800g
Mixing Speed 35/50 RPM
Mixing Time 10 min
Verify Installation and Specifications of Mesh Screen 30 mesh

Working Time (Start~End)
13:51 ~ 14:24
4,800 g
35/50 RPM
13:55 ~ 14:05
30 mesh
Suitable Unsuitable
800 g
35/50 RPM
14:06 ~ 14:08

8
Final Mixing

Jusung Ahn
08/12

Check exterior and any abnormalities of the mesh screen.

800g

Magnesium Stearate.

Under the following conditions

Put the above sifted material into the U-shaped mixer
Final Mixing

Mixing Speed

35 / 50 RPM

2 mins

Mixing Time

15/70

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POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Manufacturing Process Name

Manufacturing Unit

Manufacturing Number

Product Name

Product Code No.

(Middle Layer) Mixing

18005

200,000 tablets

Cephorin-F Tablet

P40327

Worker

Verifier
Verification Date

Record Details

Work Process

Step

Work Date

Jusang Ahn
98/13

Verify production amount of the mixture

9
Display Yield

Net Content

Container Weight

Gross Weight

No

33 kg 510 g

2 kg 120 g

36 kg 630 g

1

32 kg 650 g

3 kg 450 g

36 kg / 100 g

2

66 kg 160 g
68kg 000g
91.3 %

Total Production
Theoretical Amount
Yield= Total Production ×100 (Range: 97.0 ± 3.0%)

Theoretical Amount

※ Attach Mass Data

68/13

001 33.51 kg
002 32.65 kg
18/08/13
N 2
TOTAL 66.16 kg

Moosang Ahn
08/13

1) ☑ Packaging Verification
2) ☑ Attachment Completed
3) ☑ Storage Completed
4) ☑ Fit Verification

Managed according to In-process Intermediate Product Management Regulations (PP-2-04-003).
1) Double Vinyl Bag Packaging
2) Attach Process Verification Label on Intermediate Product Storage Container
3) Store in Intermediate Product Room 1
4) Verify Temperature and Humidity of Intermediate Product Room (20±5℃, below 65%)

10
Storage

※ Special Comments (Comment)

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

_____ Manufacturing Unit

Manufacturing Process Name

P40327

Cephorin-F Tablet

18005

200,000 tablets

(Upper, Lower Layer) Mixing

Step

Pre-Operation Check

Inspector
Inspection Date

Verifier
Verification Date

Inspection Time (Start~End)

14:44 ~ 15:01

① Confirm Scale Balance and Zero Point

Suitable □ Unsuitable

② Reconfirm Weighed Raw Material, Input and Attach Cleanliness Label Record

③ Verify Cleaning and Drying Status of Equipment and Any Abnormalities
1) SUS Container (40ℓ) (PRE-212) Suitable Unsuitable
2) Air Mixer (4) (PRM-196) Suitable Unsuitable
3) High-Speed Mixer (3) (PRM-225) Suitable Unsuitable
4) Fluid Bed Coater (GPCG60) (PRM-107) Suitable Unsuitable
5) 20㎛ Product Retaining Filter (PRE-248) Suitable Unsuitable
6) Wet Granulator (12, 24mesh) (PRM-172) Suitable Unsuitable
7) Mobile Powder Feeder (2) (PRM-172-①) Suitable Unsuitable
8) Mesh Screen (30mesh) (PRE-211) Suitable Unsuitable
9) U-Shaped Mixer (PRM-115) Suitable Unsuitable
10) Mobile Powder Feeder (PRM-115-①) Suitable Unsuitable

Jusam Ahn
08/13

This
@ [OCR] Measurement Attachment Missing

⑤ Verify Cleanliness and Cleaning Status of Workroom □ Suitable □ Unsuitable

Step

Work Process

Worker

Record Details

Verifier

Work Date

Verification Date

Verify and Record Temperature (20±5℃) and Humidity (65% or less) in Mixing Room 8 (PR227)

Temperature : 22 ℃
Humidity : 55.7 %

Man-gok Sang
08/13

08/13

Verify and record temperature (20±5℃) and humidity (below 65%) in Room 2 of Gwalim (PR36)

Inspection Details

Temperature : 21.5 ℃
Humidity : 59.8 %

Chanwook Park
08/13

JS
08/13

Temperature : 22.1 ℃
Humidity : 57.3 %

Ahn Joo-Sang
08/13

Verify and Record Temperature (20±5℃) and Humidity (65% or less) in Mixing Room 3 (PR63)

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Gaseporim F-Tablets

200,000 tablets

(Upper, Lower Layer) Mixing

Step

Work Process

Worker

Record Details

Verifier

Work Date

Verification Date

11
Preparation of Binding Solution

In a 40ℓ SUS container, sequentially add the following ingredients and prepare the binding solution with an air mixer (4)

Ethanol...24,000g
Purified Water...4,800g
Hydroxypropyl Cellulose...2,000g
Mixing Time...10±5 minutes

Visually verify that the above ingredients are dissolved

Working Time (Start~End)
15:09 ~ 15:19
24,000 g
14,800 g
2,000 g
15:11 ~ 15:18
✔ Suitable □ Not Suitable

Ahn Joo-Sang
08/13

Wind
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Chapter
Text
Main
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Working Time (Start~End)
15:21 ~ 15:30

20,000 g

60,000 g

8,000 g

110 RPM

2,000 RPM

15:25 ~ 15:30

Ahn Joo-Sang
08/13

20,000g

Sucralfate Hydrate..

12

60,000g

Adsorption

Microcrystalline Cellulose vivapur12.

8,000g

Agitator Speed

110±10 RPM

Super Impeller Speed

2,000±100 RPM

Mixing Time

5 minutes

Punglim Pharm. Co., Ltd.

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Cephorin-F Tablet

13005

200,000 tablets

(Upper, Lower Layer) Mixing

Step

Work Process

Record Details

Worker

Verifier
Verification Date

Work Date

13
Amalgamation
1

Inject the binding solution into the above mixture under the following conditions for amalgamation

Binding Solution Injection Time 2 minutes
Agitator Speed 110±10 RPM
Super Impeller Speed 2,400±100 RPM
Mixing Time 2 minutes

Working Time (Start~End)
15:30 ~ 15:39
15:30 ~ 15:32
110 RPM

2,400 RPM
15:32 ~ 15:34

Ahn Joo-Sang
08/13

After checking the mesh, dry the combined mixture under the following conditions in a fluid bed coating machine

Check mesh installation and inspect for abnormalities
Air Flow 1,500~2,800 m³/h
Inlet Temperature 65.0±10.0℃
Product Temperature 45.0±5.0℃
Drying Time 20±5 minutes

14
Drying
-

Working Time (Start-End)
15:42 ~ 16:00
☑ Suitable ☐ Not Suitable
1,500 ~ 2,800 m³/h
65.0 ℃
43.1 ℃
15:44 ~ 16:00

Chanwook Park
08/13

JS
08/13

Working Time (Start-End)
16:04 ~ 16:16
12 mesh
☑ Suitable ☐ Not Suitable
45 RPM

Jinok Park
08/13

JS
08/13

After checking the granulation mesh, inject the dried material into the wet granulator under the following conditions for granulation

15

Granulation

Check installation and specification of granulation mesh..

12 mesh

Check appearance and inspect for abnormalities of granulation mesh

Rotational Speed.

45±5RPM

2017.12.15 - Ver.08

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Form No.: FP-2-01-001-01(10)

Punglim Pharm. Co., Ltd.
Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Gasprin F Tab

18005

200,000 tablets

(Upper, Lower Layer) Mixing

Step

Work Process

Record Details

Worker

Verifier

Work Date

Verification Date

16
Preparation of Binding Solution
II

In a 40ℓ SUS container, sequentially add the following ingredients and prepare the binding solution with an air mixer (4)

Ethanol ... 24,000g
Purified Water ... 4,800g
Hydroxypropyl Cellulose ... 2,000g
Mixing Time ... 10±5 minutes

Visually verify that the above ingredients are dissolved

Working Time (Start~End)
16:23 ~ 16:33

24,000 g
4,800 g
2,000 g

16:25 ~ 16:32

✓ Suitable □ Not Suitable

Ahn Joo-Sang
08/13

Add the following ingredients into a high-speed mixer (3) for mixing

Bismuth Subcitrate Potassium ... 20,000g
Sucralfate Hydrate ... 60,000g
Microcrystalline Cellulose vivapur12 ... 8,000g

Agitator Speed ... 110±10 RPM
Super Impeller Speed ... 2,000±100 RPM
Mixing Time ... 5 minutes

Working Time (Start~End)
16:35 ~ 16:43
20,000 g
60,000 g
8,000 g
110 RPM
2,000 RPM
16:38 ~ 16:43

Ahn Joo-Sang
08/13

08/13

Form No.: FP-2-01-001-01(10)

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

18005

Cephorin-F Tablet

(Upper, Lower Layer) Mixing

P40327

200,000 tablets

Worker

Verifier

Step

Work Process

Record Items

Work Date

Verification Date

18
Amalgamation
II.

Inject the binding solution into the above mixture under the following conditions for amalgamation

Binding Solution Injection Time ... 2 minutes

Agitator Speed ... 110±10 RPM

Super Impeller Speed ... 2,400±100 RPM

Mixing Time ... 2 minutes

Working Time (Start~End)

16:43 - 16:55

16:43 - 16:45

110 RPM

2,400 RPM

16:45 - 16:47

Ahn Joo-Sang
08/13

After checking the mesh, dry the combined mixture under the following conditions in a fluid bed coating machine

Check mesh installation and inspect for abnormalities

Air Flow ... 1,500~2,800 m³/h

Inlet Temperature ... 65.0±10.0℃

Product Temperature ... 45.0±5.0℃

Drying Time ... 20±5 minutes

Working Time (Start~End)

16:58 ~ 17:17

☑ Suitable □ Not Suitable

1500 ~ 2860 m³/h

65.0 ℃

45.3 ℃

17:00 ~ 17:17

19
Drying
II

Park Chan-ok
08 / 13

JS
08/13

After checking the granulation mesh, inject the dried material into the wet granulator under the following conditions for granulation

Check installation and specification of granulation mesh... 12 mesh

Check appearance and inspect for abnormalities of granulation mesh

Rotational Speed... 45±5RPM

Working Time (Start-End)

17:18 - 17:30

12 mesh

☑ Suitable ☐ Not Suitable

45 RPM

Chanwook Park
08/13

JS
08/13

20

Granulation

|

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Product Code No.

_____ Product Name

Manufacturing Number

_____ Batch Size

Manufacturing Process Name

P40327

Cephorin-F Tablet

18005

200,000 tablets

(Upper, Lower Layer) Mixing

Step

Work Process

Worker

Verifier

Record Details

Work Date

Verification Date

After checking the mesh, apply fluid bed coating to the granulated product under the following conditions video dry

Check mesh installation and inspect for abnormalities

Air Flow 1,500~2,800 m3/h

Inlet Temperature 75.0±5.0 ℃

Product Temperature 50.0±5.0 ℃

Drying Time 40±5 minutes

Measure residual moisture content under the following conditions and end the drying process if it meets the standard range
Halogen Moisture Analyzer, 80℃, 4 minutes, 5g used
Moisture Content Standard 4.0% or less

Timer 4:00 min
Drying Temp. 80 °C
Weight 5.009 g
0...-100 %MC

0:00 min -0.56 %MC
2:00 min -1.98 %MC
-13.08.18~18:22-

Process Time (Start~End)
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Main
Bulk 48 HE 6
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[OCR]
Result; 3.60 %

21

Drying

It

Moisture Measurement

After checking the granulation mesh, inject the dried material into the mobile powder feeder (2) for granulation under the following conditions

Check installation and specification of granulation mesh... 24 mesh

Check appearance and inspect for abnormalities of granulation mesh

Rotational Speed... 45±5RPM

Working Time (Start~End)

18:24 - 18:52

24 mesh

☑ Suitable ☐ Not Suitable

45 RPM

Jinok Park
08/13

JS
08/13

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Manufacturing Process Name

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

18005

P40327

Cephorin-F Tablet

(Upper, Lower Layer) Mixing

200,000 tablets

Worker

Verifier

Step

Record Details

Work Process

Work Date

Verification Date

Working Time (Start~End)
18:55 ~ 19:36
5,600 g
16,000 g
35/50 RPM
18:58 ~ 19:08
30 mesh
Suitable Not Suitable
1,600 g
35/50 RPM
19:10 ~ 19:12

23
Final Mixing

In the U-shaped mixer, add the granulated product with the following ingredients and mix under the following conditions
Silicon Dioxide... 5,600g
Croscarmellose Sodium... 16,000g
Mixing Speed ... 35 / 50 RPM
Mixing Time ... 10 minutes
After checking the sifting mesh, sieve the following ingredients
Check installation and specification of sifting mesh... 30 mesh
Check appearance and inspect for abnormalities of sifting mesh
Magnesium Stearate... 1,600g
In the U-shaped mixer, add the above sifted product and conduct final mixing under the following conditions
Mixing Speed ... 35 / 50 RPM
Mixing Time ... 2 minutes

Ahn Joo-Sang
08/13

Check the yield of the final mixture

Ahn Joo-Sang
08/13

Net Content

No

Gross Weight

Container Weight

58 kg 700 g

3 kg 270 g

55 kg 500 g

1

24

43 kg 510 g

2

46 kg 610 g

3 kg 100 g

Process Yield

57 kg 320 g

53 kg 100 g

3 kg 620 g

45 kg 300 g

48 kg 420 g

3 kg 120 g

4

198 kg 0/0 g

Output yield

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Dongrim Pharmaceutical Co., Ltd.

Manufacturing Instruction and Record

Manufacturing Process Name

Manufacturing Unit

Manufacturing Number

Product Name

Product Code No.

compression

18005

200,000 tablets

Cephorin-F Tablet

P40327

Inspector

Inspection Date

Confirmer

Confirmation Date

Pre-Operation Check

Step

08:41 - 09:07

Inspection Time (Start~End)

① Check scale balance and zero point ☑ Conform ☐ Non-conform

② Attachment of in-process check sheet of semi-finished products, and confirmation of weight and number of containers

99 kg 000 g
Ointment Manufacturing Date: 2013.08.21

Confirmed Quantity a-

[OCR]

2

Number of Containers

Transmitter

Upper Layer

Choi Chang-seok

Recipient

66 kg 160 g

Reception Quantity

2

Number of Containers

Limit Excess

Transmitter

Increase Weight

Recipient

Reception Quantity

49 kg 0 g

2

Number of Containers

Counterpart

Transmitter

Load

③ Confirm the cleaning, drying, and absence of defects of machinery/tools
1) Rotary Compressor (5) (PRM-200)
Conform □ Non-conform
2) Geporin F Tablet Punch (AG05)
Conform □ Non-conform

Choi Chang-seok

Recipient

④ Attachment of machinery cleaning confirmation sheet
Attached □ Not Attached

⑤ Check the cleanness and cleanliness status of the workroom
Conform □ Non-conform

⑥ Attach the process confirmation sheet of the previous process before starting work
Attached □ Not Attached

SEO
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Worker

Verifier

Record Details

Step

Work Date

Verification Date

Main
Challenge 5 Room(88210) Temperature Challenge 5 Room(88210) Temperature (20+50) 6)Humidity (65% Humidity ≤65%) Confirmation and Record

Son 08/21

Temperature : 21.1 ℃
Humidity : 53.5 %

Choi Chang-seok 08/21

Inspection
Check Items

Sequence
08/21

Conform Non-conform

Choi Chang-seok
08/21

① Verification of rotary compressor (5) usage layer information and Change Part Setting status

26 Compression

Layer	1	2	3	4	5
Layer Active	■ Active	□ Inactive	■ Active	□ Inactive	■ Active	□ Inactive	■ Active	□ Inactive	■ Active
Insert Cam	A1	A2	A2	A2	X
Edge Cam	1-1	2-1	2-2	2-1	X
Dosing Cam	1-3	X	X	X	X
Filling Cam	10	X	X	X	X

Punglim Pharm. Co., Ltd.
Manufacturing Instruction and Record

Manufacturing Process Name

Manufacturing Unit

Manufacturing Number

Product Name

Product Code No.

compression

200,000 tablets

18005

Cephorin-F Tablet

P40327

Verifier

Operator Work Date

Record Details

Work Process

Step

Verification Date

Son 08/3

Choi Chang-seok
08/21

Work Time (Start ~ End)
09:15 ~ 15:02
Lower Layer: 99.000 g
Middle Layer: 66.670 g
Upper Layer: 99.010 g
9 RPM
67 Stations
39.5 KN

② Insert received mixed powder into respective hoppers
Hopper S_1... Lower Layer Quantity
Hopper S_3... Middle Layer Quantity
Hopper S_5... Upper Layer Quantity
③ Set the compressor in the following conditions
Compression Speed ... 5~10 RPM
Number of Punches ... 67 Stations
Pressure ... 35~45 KN

Tablet Specification and In-process Inspection Items, Criteria, and Frequency

Appearance: Rectangular triple-layered white

508.0mg (Upper Layer Standard Mass:508.0mg)

Compression Instruction Quantity
(1,356.0mg)

340.0mg (Middle Layer Standard Mass:340.0mg)

508.0 mg (Lower Layer Standard Mass:508.0 mg)

Length : 19.7mm
Width : 8.7mm

Size

Front: Back:

Embossing Identification

Test Compression and Every Hour Interval

6.6~7.3mm

Thickness (5T)

Test Compression and
Every Hour Interval

Standard Mass ±5.0%
1,288.2~1423.8㎎

Individual Mass
(20T)

Test Compression and
Every Hour Interval

Above 10.0kp

Hardness
(5T)

Test Compression and
30 Minute Interval

Standard Mass ±3.0%
1315.3~1396.6mg

Average Mass
(20T)

Test Compression and
Once Each Morning and Afternoon

Below 1.0%

Friability
(20T)

Seal

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40/70

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POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Manufacturing Unit

Manufacturing Process Name

Manufacturing Number

Product Code No.

Product Name

18005

compression

P40327

Cephorin-F Tablet

200,000 tablets

Worker

Record Details

Verifier

Step

Work Process

Work Date

Verification Date

③ Record and confirm verifier's confirmation after 3 rotations test compression results

Thickness (mm)
(67T)

Friability (%)
(20T)

Seal

Worker

Verifier

Start Date

Time

Hardness (kp)
(67T)

Individual Mass (mg)
(67T)

Average Mass (mg)
(67T)

0.10

Choi Chang-seok

23.61

6.98

| Son Hyo-young

| 08/21

09:15

i. 320 v i. 383

i. 258

Moon

Notice

④ Record process inspection results in the Main processing progress and manufacturing process inspection record Main

① Signature of process control manager on remaining quantity confirmation form
② Attachment of remaining quantity confirmation form
③ Storage of waste during process

Remaining: 18,650 g
☑ Attachment Completed
☑ Create Transfer Record

Choi Chang-seok
08/21

27
Remaining
Processing

Verify the Production Volume of the Compressed Tablets

Choi Chang-seok
08/21

Son
Ya/Na

Container Weight

Net Content

No

is
[OCR]

[OCR]Document

64 kg 240 g

3 kg 510 g

1

[OCR]Main[OCR]
[OCR]

58 kg 140 g

3 kg 560 g

②

65 kg 740 g

3 kg 290 g

69 kg 030 g

28

55 kg 400 g

3 kg 480 g

58 kg 870 g

Pro
gress
Yield

243 kg 520 g

179,679 tablets

194,901 tablets

194,588 tablets

194,881 tablets

Total Production Volume

Total Production Quantity = Total Production Volume + Average Mass of Compressed Tablets

Theoretical Quantity for Upper Layer = Quantity of Mixed Material for Upper Layer ÷ Instruction Quantity for Upper Layer Compression

Theoretical Quantity for Middle Layer = Quantity of Mixed Material for Middle Layer ÷ Instruction Quantity for Middle Layer Compression

Theoretical Quantity for Lower Layer = Quantity of Mixed Material for Lower Layer + Instruction Quantity for Lower Layer Compression

Manufacturing Instruction and Record

Manufacturing Unit

Manufacturing Process Name

Manufacturing Number

Product Name

Product Code No.

18oct5

compression

200,000 tablets

Cephorin-F Tablet

P40327

Worker

Verifier

Record Details

Work Process

Step

Work Date

Verification Date

1.355.3 mg
92.3 %
Lim
Se
Moon

Choi Chang-seok 08/21

Average Mass of Compressed Tablets

Yield = Total Production Quantity x 100 (Range: 90.0 ± 10.0%)

Theoretical Quantity

※ When calculating yield, the theoretical quantity is based on the minimum theoretical quantity among upper, middle, and lower layers.

※ When calculating yield, the theoretical quantity is based on the

minimum theoretical quantity among upper, middle, and lower layers.

※ Attach Mass Data

18/08/21
001 64.24 kg
002 58.14 kg
003 65.74 kg
004 55.40 kg

18/08/21
N 4
TOTAL 243.52 kg

This document is a copy issued by Pulrim Pharmaceutical Co., Ltd.

1) Packaging Confirmation
2) Attachment Completed
3) Storage Completed
4) Suitability Confirmation

Choi Chang-seok 08/21

Son 08/21

Managed according to In-Process Semi-Product Management Regulations (PP-2-04-003).
1) Double plastic bag packaging
2) Attach Process Verification Tag to Semi-Product Storage Container
3) Store in Semi-Product Room 7
4) Confirm Temperature and Humidity in Semi-Product Room (20±5℃, below 65%)

29
Stor
age

※ Special Comments (Comment)

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Poonglim Pharmaceutical Co., Ltd.

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Cephorin-F Tablet

18005

Coating

200,000 tablets

Work Process

Step

Record Details

Operator
Date of Work

Verifier

Verification Date

30
Preparation of Coating Solution

In a 100L SUS container, add the following raw materials in order and prepare the coating solution using air mixer (6)
Purified Water ... 51,200g
Opadry Green AMB 80W610017... 13,312g
Dispersion Time ... Within 1 Hour

Visually confirm whether the above raw materials are dissolved.
Prepare the coating solution by passing the stirred coating solution through the sieve net
Confirm installation and specifications of sieve net ... 80 mesh
Inspect appearance and abnormalities of sieve net
Use the film coating solution within 24 hours after manufacture

Working Hours (Start~End)
08:43 - 09:40
51,200g
17,313g
08:44 - 09:40
□ Suitable □ Unsuitable
80 mesh
☑ Suitable □ Unsuitable

Lee Cheon-woo
08/22

Kim .
08/22

Perform coating with Film Coating Machine (4) under the following conditions
Spray Nozzle Orifice Size Ø 1.2 mm
R.P.M. 50.0±20.0 RPM
Atom Air 0.3 ± 0.1 MPa

Working Hours (Start~End)
09:56 - 14:23
ø 1.2 mm
35~67 RPM
0.3 MPa
0.7 MPa
0.5 MPa
3.5~5 RPM
28 cm
74~88 °c
40~49 °c
14:11 - 14:23

Lee Cheon-woo
08/22

ø 1.2 mm

31

Main Air

0.6 ± 0.1 MPa

Coating

Cylinder Air

0.5 ± 0.1 MPa

Pan R.P.M

6.0 ± 3.0 RPM

Spraying Distance

23 ± 5 cm

Inlet Temp.

70.0 ± 20℃

Outlet Temp. ...

50.0 ± 10℃

After completing the spray, rotate the pan for about 20 ± 10 minutes to dry the coated tablets

Dongrim Pharmaceutical Co., Ltd.

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Cephorin-F Tablet

18005

Coating

200,000 tablets

Step

Work Process

Record Details

Worker

Verifier

Work Date

Verification Date

Verify and Record Physical Characteristics, Average Mass after Coating Completion

Lee Sa-yoon
08/22

Lim Tae-yoon
08/23

Appearance

Green oblong film-coated layered tablet

5i0)72 {0122 MEK
[OCR] [OCR]
Lim

Average Mass
(100T)

Standard Mass

1,422.56mg

Standard Mass± 3.0%
1,379.9~1,465.2mg

[ATE 2818-08-22 [OCR]Issued
※ Mass Data Attachment
[OCR][OCR] [OCR] [OCR] Document
[OCR]
MI. 3 Main
HO, 18 Four
WT 14,0581 =
One
WT 14.815 =
I ia
[OCR] Main
TOTAL
MA 14.128 = a= to
[OCR]
MIN 13.945 = about
[OCR] [OCR] Weapon
[OCR] [OCR] [OCR]
[OCR] [OCR] [OCR] Lim

[OCR]
Wind
R 3.174 5
[OCR]Main
Document

Main
Verify Production Volume of Coated Tablets.

NO

Gross Weight

Container Weight

Net Content

1

65 kg 480 g

3 kg 170 g

62 kg 310 g

32

②

66 kg 180 g

3 kg 110 g

65 kg 070 g

Process Yield

③

65 kg 200 g

3 kg 200 g

61 kg 400 g

4

68 kg 100 g

3 kg 470 g

64 kg 630 g

251 kg 910 g
179,579 tablets

Total Production Volume

Total Production Quantity = Input Quantity - Process Test Quantity

2017.12.15 -- Ver.08

Form No.: FP-2-01-001-01(10)

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Cephorin-F Tablet

18005

Selection

200,000 tablets

Step

Work Process

Operator
Date of Work

Record Details

Verifier
Verification Date

Manual sorter usage

Yes, it is available.

No, it is not available.

Inspection Time (Start~End)

09:20 - 09:35

① Check scale leveling and zero setting

Conform Non-conform

② Attachment of in-process check sheet of semi-finished products, and confirmation of weight and number of containers

251 kg 800 g

Transmitter

Number of Containers

4

Received quantity

Recipient

Ha Eunji

③ Check for washing, drying, and any abnormalities of machinery
1) Sorter(2)(PRM-163),Metal detector(2)(PRM-167)
□Conform□Non-conform☑N/A
2) Sorter(4)(PRM-223),Metal detector(4)(PRM-224)
☑Conform□Non-conform□N/A

38

Work

Before

④ Attach equipment cleaning confirmation sheet
☑Attached □ Not attached
⑤ Check the cleanliness and cleaning status of the workroom
☑Conform□Non-conform
⑥ Attach previous process confirmation sheet before starting work
☑Attached □ Not attached

Inspection

Ion paper
08/23

08/23

Check and record temperature (20±5℃) and humidity (under 65%) in Selection Room 4 (PRD∞)

Temperature: 20.5 ℃
Humidity: 56.1 %

Conduct performance check of metal detector 3 times before starting work (indicate as O,X)

Yeo Eunji
08123

Gi
08123

39

Time/Count
Sample

09 : 37

13 : 03

Metal detector

1st round

2nd round

3rd round

1st round

2nd round

3rd round

Blank

Fe

Performance check

Non-Fe

AISI 316

0

POONG LIM PHARMACEUTICAL CO., LTD

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Cephorin-F Tablet

18005

200,000 tablets

Selection

Step

Work Process

Operator
Date of Work

Verifier
Verification Date

Record Details

40
Manual selection

Proceed with manual selection after completing the performance check of the metal detector.

Working time (Start~End)
09:50 - 17:30

Incomplete 08/23

Product 08123
Product

Selection defect

Metal foreign object

Others

12.560 g

12 . g

10 g

Seal the selected portion immediately after selection and manage defective selections.

41
Residual handling

① Approval from the process management officer
② Attach residual confirmation sheet
③ Store waste during process

Total: 12,570 g
Defective: 8,960 tablets
□ Attachment completed
■ Draft export history

Ha Unji
8/23

Ha Unji
8/23

Accurately confirm the production quantity of selected tablets

Container Weight

No

Gross Weight

Net Content

66 kg 600 g
66 kg 420 g

3 kg 480 g

63 kg 120 g

3 kg 300 g

63 kg 120 g

66 kg 450 g

3 kg 330 g

63 kg 120 g

53 kg 180 g

3 kg 310 g

42

4 kg 810 g

Pro
gress
Yield

5

kg g

kg g

kg g

6

kg g

kg g

kg
g

239 kg 230 g
110,539 tablets
119,499 tablets
95.0 %

Total production

Lee Eunji
08/23

08/23

Total production quantity = Theoretical quantity - Defective quantity

Theoretical quantity = Coated tablet received quantity - Test sample quantity

Total production quantity

Yield =

× 100 (Range: 95.0 ± 5.0%)

Theoretical quantity

Manufacturing Instruction and Record

Product Code No.

Product Name

Manufacturing Number

Manufacturing Unit

Manufacturing Process Name

P40327

Cephorin-F Tablet

1800T

200,000 tablets

Bulk packaging

Step

Inspector

Inspection date

Verifier

Verification date

Work

Before

Inspection

Inspection time (Start-End)

09:00 - 09:16

Kim Jeonggi

Check if it matches the material requested quantity

Match □ Mismatch

1

Material name

Requested quantity

Reception Quantity

Recipient

Final box (Bulk)

12 EA

12 EA

Kim Jeonggi

Scale leveling and zero point check
Confirmation of attachment of semi-product selection sheet and weight, number of containers
Carrier YunXX 4 containers Intake amount 239kg 230g
Learner Kim Jung-gi
Cleanliness and cleaning status of the workshop checked
Conforming Non-conforming
Attachment of the previous process selection sheet before starting work
Attached Not attached

Scale leveling and zero point check

2

3

Verifier Sign off

Initial Check

Number of Containers

4

Intake amount

239 kg 230 g

Recipient

Kim Jeonggi

4

Work Process

Step

Record Details

Worker Verifier
Work Date Verification Date

1) For product shipment, pack the semi-products in vinyl packaging into the final box
2) Attach the product shipment label to the final box

44
Pack
aging

15,000 pcs x 11 BOX
5,539 pcs x 1 BOX

Kim Jung-gi
08/27

45
08/27

45
Product Dispatch

1) Issue product inbound and outbound slips
2) Transfer products to the production support team

Handover Complete

Recipient: Tae Man-wi

Kim Jung-gi
08/28

08/28

Remarks(Comment)

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